SECTION 3 · SUBSECTION 3.8
WEB API AND JOB CONTROL

3.8.1 INTRODUCTION.

The web API is a thin scheduling and status surface. It validates a submission, refuses duplicate active work, measures the repository before accepting it, creates a job record, and hands the job to either Redis Queue or a local thread pool. Every response is derived from the job store; the API holds no analysis result in memory of its own. The one idea the reader must leave with is that acceptance and execution are separate, so a 202 means scheduled, not finished (figure 3.8-1).

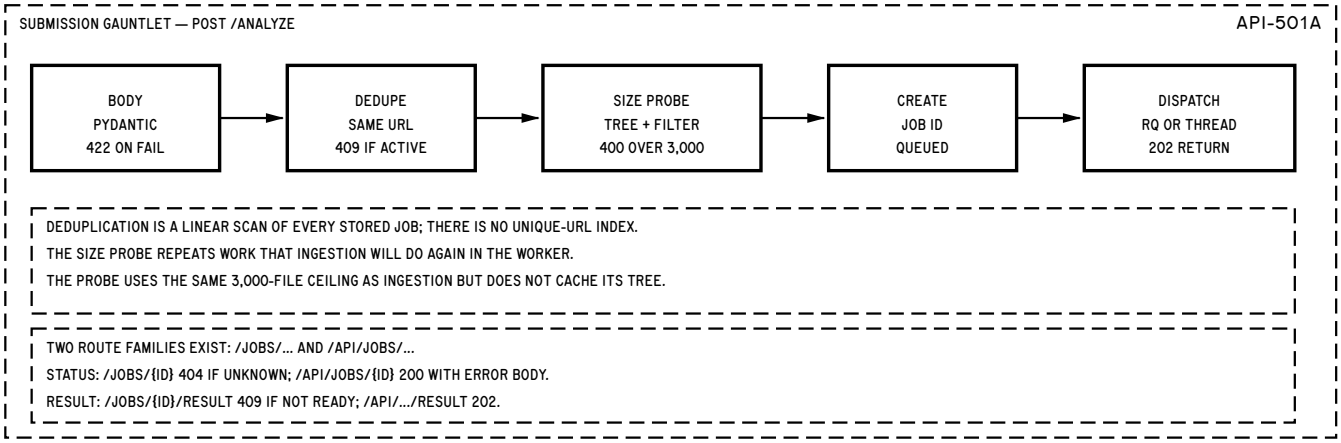


Figure 3.8-1. Submission Validation and Dual Route Behaviour

3.8.2 FUNCTIONAL DESCRIPTION.

The request model accepts *repo_url*, *max_explanations* (0 to 100), *skip_llm*, *github_token*, *incremental* and an optional *options* dictionary that can override the four execution fields. The URL validator requires the substring *github.com* and at least three path segments; it does not reject a non-GitHub host that contains the string. After deduplication, the route fetches repository metadata and the full tree to count filtered files, rejecting more than 3,000. It then creates the record and dispatches it. Status routes read the record; result routes return it or an error.

3.8.2.1 Synchronous Size Probe.

The probe runs inside the request, not in the worker. A large repository therefore delays the 202 response by the duration of two GitHub calls plus filtering, and a GitHub outage becomes a 400 with the text "Failed to validate repository size". The probe is a guard against wasting a worker slot, not a security boundary.

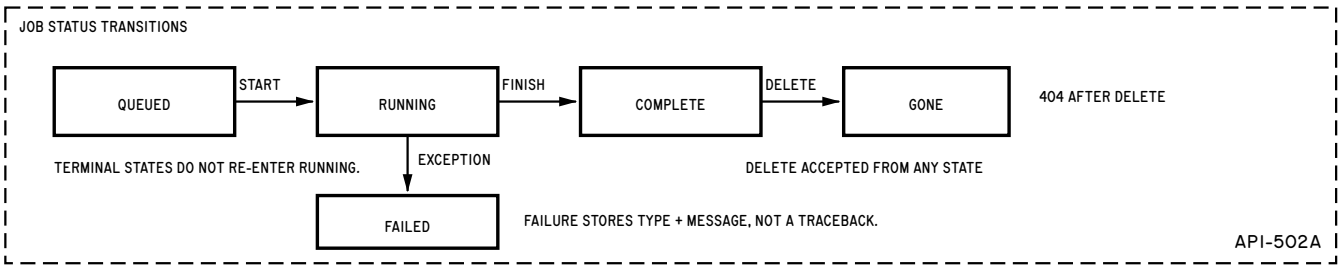


Figure 3.8-2. Stored Job Status Transitions

3.8.3 PERFORMANCE AND DESIGN DATA.

Table 3.8-1. HTTP Surface

Route	Success	Failure
POST /analyze	202 + job id	400, 409, 422
GET /jobs/{id}	200 record	404
GET /api/jobs/{id}	200 record	200 error body
GET /jobs/{id}/result	200 result	404, 409, 500
GET /api/jobs/{id}/result	200 result	202, 404, 500
GET /jobs	200 list	—
DELETE /jobs/{id}	204	404
GET /health	200 status	—

CAUTION

CORS allows every origin with credentials enabled, while the frontend also defines a strict content policy. The two settings serve different layers and do not cancel each other. Any site can call this API from a browser; do not place it behind assumptions of origin isolation.

API CHECK	OPERATOR
1 GET - /health, read queue mode	
2 POST - a small public repository	
√ 202 - job id returned	
3 POLL - /api/jobs/{id} advances	
4 FETCH - result after complete	

API